HOW CLUED UP ARE LLMS?

Evaluating multi-step deductive reasoning in a text-based game environment

01 INTRODUCTION



LLMs are increasingly deployed as agents in interactive environments requiring structured reasoning. However, existing benchmarks evaluate reasoning in isolation, not over extended, stateful interactions. We implement a text-based multi-agent Clue environment as a testbed for evaluating multi-step deductive reasoning, with six agents drawn from GPT-4o-mini and Gemini-2.5-Flash. We further investigate whether fine-tuning on structured logic puzzles (Mind Benders) transfers to improved in-game reasoning and gameplay.

02 RESEARCH QUESTIONS

RQ1: Can LLM agents sustain logically consistent deductive reasoning across extended, multi-turn interactions in a structured game environment?

RQ2: Does fine-tuning on deductive reasoning tasks improve agentic performance in a related environment?

RQ3: How does knowledge accumulation relate to reasoning quality in fine-tuned versus base model variants?



03 THE CLUE ENVIRONMENT

A modified text-based version of Clue (no board/dice) where LLM agents must:

1. Narrow candidate hypotheses through iterative elimination
2. Maintain logical consistency across an extended interaction
3. Integrate asymmetric information sources
4. Determine when sufficient evidence exists for an accusation



04 EXPERIMENTAL DESIGN

We implement LLM-based agents that interact with the Clue environment through structured prompts and formatted responses. Our design separates game logic from agent reasoning.



Players

2 model families, 4 variants:

GPT-4o-mini (base + FT)

Gemini-2.5-Flash (base + FT)

Temperature 0.7 · 6 agents per game

Fine-Tuning

Models are fine-tuned on mind bender logic puzzles (50 training, 10 validation)

GPT-4o-mini FT: 88% train, 87% val

Gemini FT: 98% train, 82% val

Experiments

2 Simulations:

1. Baseline: 6 games, 3x GPT-4o-mini, 3x Gemini-2.5-Flash
2. Fine-tuned: 12 games, 2x GPT-4o-mini, 2x Gemini-2.5-Flash, 1 x FT 4o, 1x FT 2.5-Flash

Game Setup

EVALUATION METRICS

- FINISHING RANK
- ACCUSATION ACCURACY (0-3)
- DEDUCATION QUALITY

CONTROL

- 30 ROUND LIMIT
- ROTATING STARTING POSITIONS
- KNOWLEDGE ACCUMULATION TRACKED

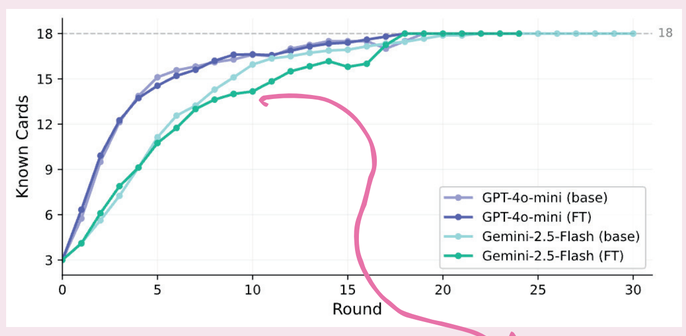
TURN STRUCTURE

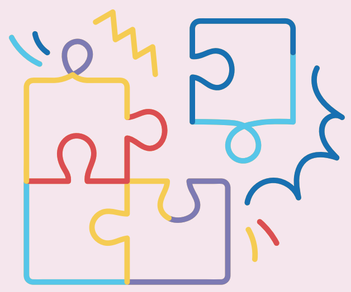
3 REASONING PHASES PER TURN:

DEDUCTION → ACTION → SHOWN CARD

(3 RE-PROMPTS ALLOWED ON A FAILED PARSE)

05 RESULTS





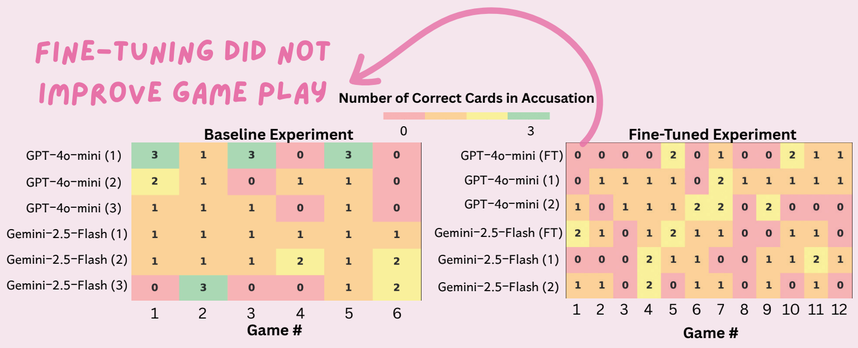
GPT-4O-MINI MODELS HAVE STRONGEST PERFORMANCE

STRONG MAJORITY OF CARDS KNOWN IN 10 ROUNDS

FINE-TUNING DID NOT IMPROVE GAME PLAY

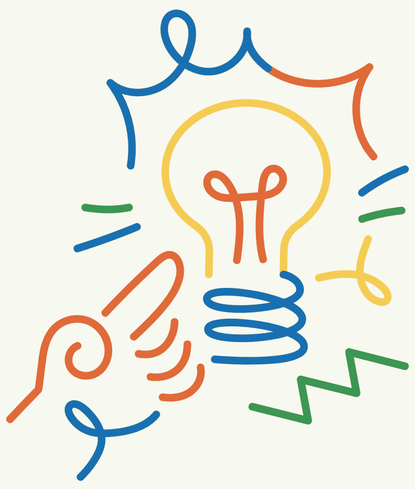
	OUTCOME			REASONING		RELIABILITY
	Wins	Rank	Acc.	Ded.	✗ Ded.	Fallbacks
GPT-4o-mini (base)	**5**	**2.96**	**0.29**	10.1	2.7	**0.3**
GPT-4o-mini (FT)	3	4.00	0.19	**13.1**	3.0	4.8
Gemini-2.5-Flash (base)	2	3.92	0.24	9.1	2.9	8.1
Gemini-2.5-Flash (FT)	2	3.25	0.28	6.9	**2.2**	3.3

TABLE 1. Performance summary across 12 fine-tuned experiment games.
Outcome: Wins (1st place), mean finishing position (Rank; 1=best, 6=worst), and normalized accusation accuracy (0-1).
Reasoning: Average number of correct and incorrect deductions per game.
Reliability: Average fallback (reprompt) failures per game.



06 DISCUSSION & CONCLUSION

- 4/18 Wins: sustained deduction is difficult for LLMs
- FT ≠ better: logic-puzzle training did not transfer to gameplay
- Volume ≠ precision: most deductions led to the worst accusations
- Agent mix matters: base models regressed alongside FT opponents
- Next steps: Introducing tool-augmented deductive reasoning



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